1. K-Nearest Neighbors (KNN)

1. What is the main idea behind KNN classification?
A. Learn a sequence of decision rules
B. Assign a class based on nearby training examples
C. Fit a straight line to the data
D. Build multiple decision trees

2. In KNN, what does K represent?
A. The number of input features
B. The number of classes
C. The number of neighbors considered
D. The number of training iterations

3. A KNN classifier has n_neighbors=5. How is a new sample usually classified?
A. By the class of the first training sample
B. By a vote among its five nearest training samples
C. By averaging all training labels
D. By selecting five random classes

4. Which Python class is used for KNN classification in scikit-learn?
A. KMeans
B. KNeighborsClassifier
C. NearestCentroidRegressor
D. NeighborTree

5. Which import statement is appropriate for KNN classification?
A. from sklearn.neighbors import KNeighborsClassifier
B. from sklearn.tree import KNeighborsClassifier
C. from sklearn.linear_model import KNeighborsClassifier
D. from sklearn.ensemble import KNeighborsClassifier

6. Which method trains a KNN model in scikit-learn?
A. model.learn(X_train)
B. model.fit(X_train, y_train)
C. model.predict(X_train, y_train)
D. model.train(y_train)

7. Which method predicts labels for new samples?
A. model.fit(X_test)
B. model.score(y_test)
C. model.predict(X_test)
D. model.transform(y_test)

8. Why is feature scaling often important for KNN?
A. KNN requires features to be integers
B. Features with larger numerical ranges may dominate distance calculations
C. Scaling increases the number of classes
D. Scaling removes the need for training data

9. Consider age ranging from 18 to 80 and annual income ranging from 10,000 to 2,000,000. What may happen if KNN uses these features without scaling?
A. Age will automatically receive more weight
B. B. **Income may dominate the distance calculation**
C. KNN will ignore both features
D. The model will become a decision tree

10. Which tool is commonly used to standardize features before KNN?
A. StandardScaler
B. LabelEncoder on the target only
C. DecisionTreeClassifier
D. confusion_matrix

11. What is a sensible way to combine scaling and KNN in Python?
A. Pipeline([("scaler", StandardScaler()), ("knn", KNeighborsClassifier())])
B. KNeighborsClassifier(StandardScaler())
C. StandardScaler(KNeighborsClassifier())
D. Pipeline([("knn", KNeighborsClassifier()), ("scaler", StandardScaler())])

12. Why should a scaler be fitted only on the training data?
A. To reduce the number of classes
B. To prevent information from the test set entering model preparation
C. To make KNN use Euclidean distance
D. To remove all missing values

13. What generally happens when K is set to 1?
A. The classifier considers the single closest training example
B. The classifier uses all training examples
C. The classifier becomes a regression model
D. The classifier ignores distance

14. A very small K can make KNN particularly sensitive to what?
A. Feature names
B. Noise and unusual training examples
C. The order of Python imports
D. The number of CPU cores only

15. A very large K may cause which problem?
A. Predictions may become overly smooth and miss local patterns
B. The model memorizes every training label perfectly
C. Distance calculations stop working
D. The number of input features becomes zero

16. What is the purpose of trying several K values with cross-validation?
A. To choose a K that performs well on unseen data
B. To change the target labels
C. To avoid using a test set entirely
D. To make all predictions identical

17. For a binary classification problem, why might someone initially try an odd K?
A. It guarantees perfect accuracy
B. It can reduce voting ties
C. It eliminates the need for scaling
D. It makes KNN train faster in every case

18. What does weights="distance" do in KNeighborsClassifier?
A. Gives closer neighbors greater influence
B. Gives every neighbor zero influence
C. Weights features according to their names
D. Converts classification into regression

19. How does weights="uniform" treat the selected neighbors?
A. All selected neighbors contribute equally to the vote
B. Only the closest neighbor votes
C. More distant neighbors always vote twice
D. Each feature receives the same coefficient

20. Which distance measure is commonly associated with straight-line distance between points?
A**. Euclidean distance**
B. Gini impurity
C. Log loss
D. Information gain

21. Which distance measure sums absolute differences across features?
A. Manhattan distance
B. Euclidean distance
C. Entropy
D. Precision

22. In scikit-learn KNN, what does metric="manhattan" change?
A. The distance used to find neighbors
B. The number of target classes
C. The training labels
D. The evaluation metric returned by score

23. What happens when n_neighbors is larger than the number of available training samples?
A. Prediction can fail because there are too few samples
B. KNN creates new training samples automatically
C. The model switches to Naive Bayes
D. KNN silently removes all features

24. Which type of task is KNeighborsRegressor designed for?
A. Predicting a continuous value
B. Grouping data without labels
C. Predicting only text labels
D. Selecting important features

25. How does basic KNN regression predict a value?
A. It combines target values of nearby training samples
B. It chooses the most common class label
C. It fits one global linear equation
D. It counts the number of decision tree leaves

26. Which is an appropriate KNN regression problem?
A. Predicting house price from features of similar houses
B. Assigning an email to spam or not spam
C. Identifying whether an image contains a cat
D. Classifying a tumor as benign or malignant

27. Which is an appropriate KNN classification problem?
A. Predicting a patient's exact blood pressure
B. Predicting whether a patient belongs to a risk category
C. Predicting a product's selling price
D. Predicting tomorrow's temperature

28. Why can KNN prediction become slow on a very large training set?
A. It may need to search many stored training samples for neighbors
B. It retrains a neural network for each sample
C. It builds a new random forest for each prediction
D. It must calculate entropy for every feature

29. What does it mean that KNN is an instance-based method?
A. It uses stored training examples when making predictions
B. It creates one rule for every class and discards the data
C. It requires exactly one feature
D. It trains only on images

30. How can irrelevant features affect KNN?
A. They can distort which samples appear close
B. They are always ignored automatically
C. They increase the number of labels
D. They turn continuous targets into categories

31. What is the "curse of dimensionality" in the context of KNN?
A. Distances may become less informative as the number of features grows
B. Python cannot store more than three features
C. KNN requires one neighbor per feature
D. High-dimensional data always gives perfect predictions

32. Two records have identical feature values but different class labels. What difficulty can this create for KNN?
A. The nearby evidence conflicts
B. KNN cannot store either record
C. The number of features doubles
D. Distance becomes negative

33. Which scikit-learn call returns class probabilities from a fitted KNN classifier?
A. model.predict_proba(X_test)
B. model.predict_probability(y_test)
C. model.fit_proba(X_test)
D. model.class_probabilities()

34. In a three-neighbor uniform-voting KNN model, two neighbors belong to class A and one to class B. What is the predicted class?
A. A
B. B
C. Both classes are always returned
D. The prediction cannot be made

35. What is a reasonable first step if one feature is measured in millimeters and another in kilometers?
A. Scale the features appropriately
B. Increase K to the number of training samples
C. Remove the target column
D. Use training accuracy as the only evaluation

36. What does train_test_split(X, y, test_size=0.2) commonly do?
A. Reserves approximately 20% of samples for testing
B. Uses 20 features for testing
C. Removes 20% of the classes
D. Sets K to 0.2

37. Why should test samples remain separate while selecting K?
A. To keep the final test evaluation independent
B. To make the training set smaller for no reason
C. To force Euclidean distance
D. To prevent the use of cross-validation

38. Which metric is useful when classifying a rare disease and missing a positive case is especially costly?
A. Recall for the positive class
B. Training set size alone

C. Number of features alone
D. Prediction time alone

39. What does model.score(X_test, y_test) return for KNeighborsClassifier by default?
A. Classification accuracy
B. Mean squared error
C. Number of neighbors
D. Euclidean distance

40. Why can accuracy be misleading on a highly imbalanced classification dataset?
A. Predicting the majority class can produce high accuracy while missing the minority class
B. Accuracy always becomes negative
C. KNN cannot use imbalanced data
D. Accuracy is defined only for regression

41. A dataset contains a categorical feature such as blood type. What should you consider before applying standard distance-based KNN?
A. How to encode the category and whether the resulting distance is meaningful
B. Converting the target to a floating-point value
C. Setting every blood type to the same number
D. Increasing the number of test samples

42. Why is assigning arbitrary numbers such as 1, 2, and 3 to unordered categories potentially problematic for KNN?
A. The numbers imply distances and ordering that may not exist
B. KNN accepts only strings
C. Numeric features are ignored
D. The target must contain three classes

43. What is the likely effect of adding many unrelated random features to a KNN model?
A. Neighbor selection may become less reliable
B. The model must always improve
C. K automatically becomes 1
D. Scaling becomes impossible

44. What does GridSearchCV help you do with a KNN pipeline?
A. Evaluate candidate hyperparameters using cross-validation
B. Create target labels from scratch
C. Replace the test set with predictions
D. Draw a decision tree

45. Which parameter would you vary to study how many neighbors influence predictions?
A. n_neighbors
B. random_state
C. max_depth
D. C

46. What is a limitation of applying ordinary KNN directly to raw, high-resolution images?
A. Pixel-space distance may not match meaningful visual similarity
B. Images cannot be converted into arrays
C. KNN accepts only two input features
D. KNN always returns continuous values

47. If closer neighbors should matter more, which setting is most relevant?
A. weights="distance"
B. weights="uniform"
C. max_depth=3
D. criterion=“entropy"

48. When is KNN especially intuitive as a baseline?

A. When similar feature profiles are expected to have similar outcomes
B. When no labeled training examples are available
C. When every prediction must be a linear equation
D. When the target has no relationship with the features

49. In model.fit(X_train, y_train), what does X_train contain?
A. Training input features
B. Predicted probabilities
C. Test labels
D. Evaluation scores

50. In model.fit(X_train, y_train), what does y_train contain?
A. Training target values or class labels
B. Feature scaling parameters only
C. Test predictions
D. Distance metrics